# Benjamin Antonio Correia

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## EDUCATION

**University of Massachusetts Amherst** **Amherst, MA**

*BS in Informatics, Data Science Concentration | Minor in Computer Science (Major GPA: 3.7)* *Graduation Date: December 2025*

- Relevant Coursework: Data Science, Data Analytics, Statistics, Data Structures, Algorithms, Calculus, Database Management, Linear Algebra, Computer Systems, Business Intelligence, Machine Learning

## EXPERIENCE

**Engineering Development Program (EDP) | Data Engineer** **Hartford, CT**

*Travelers Insurance | Data Management and Governance Circle* *Feb 2026 - Present*

- Maintain Python ETL pipelines processing tens to hundreds of thousands of rows daily for the data asset inventory and Alation.
- Developed an event-driven Python pipeline that listens for events from Kafka and loads them into Alation through its APIs.
- Developed a daily Python batch pipeline that extracts application ID data from Snowflake via SQL and loads it into MongoDB.
- Migrated codebases from on-premises systems to Databricks, working in Python.
- Compared performance metrics of PDQ, Travelers' Great Expectations-based data quality tool, against base Great Expectations.
- Collaborate with cross-functional teams in an Agile environment using Git, CI/CD, and Docker, and use Claude Code to speed up development, verifying its output before use.

**Business Insights & Analytics Leadership Development Program | Decision Science Intern** **Hartford, CT**

*Travelers Insurance* *June 2025 - Aug 2025*

- Performed a detailed analysis of the influence of insurance quote turnaround time on conversion rate, utilizing data visualization and regression analysis to identify key trends and improve decision-making and task-queuing processes.
- Utilized Excel, SQL, R, Python, and other analytical tools to mine and analyze data, generating actionable insights.
- Supported business partners by providing reporting, dashboards, and analytical solutions to help them achieve their objectives.
- Communicated with business partners to clarify business goals, ensuring alignment and understanding to effectively tailor solutions that meet their needs.
- Assisted in the process of eliciting, analyzing, specifying, communicating, and validating business intelligence requirements while identifying project issues that require escalation.
- Engaged in comprehensive, industry-leading training focused on various technical subjects, leadership development, the insurance industry, and professional skills.

**Machine Learning & Cybersecurity Intern** **Waltham, MA**

*Dassault Systémes | Solidworks* *May 2024 - Aug 2024*

- Researched and implemented machine learning techniques to detect network intrusions by monitoring network traffic.
- Transformed and refined large datasets using Pandas, Numpy, and Scikit-learn for improved model performance.
- Trained highly precise tree-based models for threat detection based on known attack signatures, achieving 99.9% accuracy with real-time detection capabilities.
- Developed a feed-forward neural network from scratch using Numpy and implemented an autoencoder in PyTorch to detect anomalous network traffic, potentially indicating new attack vectors.

**Data Science Undergraduate Course Assistant (TA)** **Amherst, MA**

*University of Massachusetts Amherst* *Sept 2024 - May 2025*

- Assist with in-class activities and maintain awareness of course content by attending lectures.
- Grade class assignments, labs, and exams, ensuring consistent and accurate evaluation.
- Lead lab sessions and facilitate discussions to reinforce key concepts in statistics, data analytics, and machine learning.
- Monitor Piazza to provide timely assistance and answer student questions about coursework.
- Support coursework, which was entirely conducted in the R programming language.

## PROJECTS

**Analyzing Credit Risk Using Classification Models**

- Utilized RMarkdown to report on logistic regression and random forest credit risk models that predict loan defaults.
- Assessed and compared the performance of multiple models using confusion matrix-based evaluation methods.
- Identified key predictors of loan defaults and employed subject matter expertise to explain the underlying reasons.

## SKILLS

**Skills/Competencies:** Data Analytics | Predictive Analysis | Machine Learning | Anomaly Detection | AI | Databases | Data Cleaning | Data Preprocessing | Data Science | Statistics | Statistical Modeling | Business Intelligence | Algorithms | Data Structures | Data Visualization | Data Mining | Teaching | Insurance | Financial Services

**Languages/Libraries:** Python | Pandas | Scikit-Learn | Matplotlib | Numpy | PyTorch | SQL | R | RMarkdown | HTML | CSS

**Tools:** Microsoft Office | VSCode | Jupyter Notebooks | Linux | Rstudio | SQLite Studio | Tableau | Git | Github | MicroStrategy | Qlik Sense | Snowflake | Databricks | Spark | AWS | Kafka | Docker | CI/CD | Claude Code | Alation | MongoDB | Great Expectations